Detecting Jurisdiction Compliance Gaps in Process Plant Engineering with Regulatory RAG

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Abstract

Engineering calculations in Korean process plant projects span seven disciplines, but conventional review remains discipline-siloed and rarely surfaces Korean-jurisdiction obligations beyond international code compliance. This paper presents an AI verification platform that combines seven standards-based calculation engines, a four-layer verification stack, and a local KOSHA regulatory RAG layer built from 1,327 guide documents and 3,102 statutory-provision rows, yielding 16,174 searchable entries in a SQLite FTS5 index. Regulatory grounding is generated on-premises with `qwen3:4b` served through Ollama. Across 220 synthetic golden cases, all calculations pass end-to-end implementation verification. In a 60-scenario cross-discipline ablation, enabling the validator increases blocking from 0/60 to 26/60 (+0.4333), with 22 of 26 detections concentrated in piping-vessel nozzle-margin coupling. In a 50-query curated retrieval benchmark, enhanced retrieval improves Recall@1 from 0.4400 to 0.7400 and MRR@10 from 0.5744 to 0.7933 over plain BM25; Recall@3 and Recall@5 each improve by +0.12 but are not statistically distinguishable from zero on this benchmark. Three case studies show that the KOSHA RAG layer surfaces Korean-jurisdiction obligations, including Article 256 of the *Rules on Occupational Safety and Health Standards* (산업안전보건기준에 관한 규칙), even when ASME/API calculations clear without flag. We position the system as a **compliance co-pilot** for HAZOP, RBI, and digital-twin workflows and formalise the **jurisdiction compliance gap** between international code compliance and Korean PSM regulatory fulfilment.

Keywords: KOSHA RAG · process safety management · process plant engineering · multi-discipline verification · jurisdiction compliance gap · large language model · regulatory compliance

1. Introduction

Engineering calculation and compliance verification in Korean petrochemical and process plant projects span more than seven engineering disciplines — piping, pressure vessels (static equipment), rotating machinery, electrical systems, instrumentation, structural steel, and civil structures — across a broad lifecycle that includes process design, detailed engineering, construction and commissioning, and ongoing operation and inspection. Whether verifying that a new piping system meets minimum wall-thickness requirements under ASME B31.3 or assessing the remaining service life of an operating pressure vessel under API 579-1, the underlying calculation workflows share the same structural problem: discipline-specific tools and review procedures create a systematic gap in which (a) cross-discipline coupling hazards are not captured and (b) jurisdictional regulatory obligations beyond the international codes are not surfaced. Excessive piping nozzle loads can drive rotating equipment vibration; civil foundation settlement can propagate into rotating equipment alignment failure; neither is detectable by single-domain calculations alone. Likewise, an SA-106 carbon-steel line in sour-chloride service can pass ASME B31.3 minimum-wall and API 570 inspection-interval checks while still being non-compliant with Article 256 of the Korean *Rules on Occupational Safety and Health Standards*.

Korean plants subject to the *Chemical Substances Control Act* and the *Occupational Safety and Health Act* must comply with the Process Safety Management (PSM) framework administered by the Korea Occupational Safety and Health Agency (KOSHA), which imposes jurisdiction-specific procedural, documentation, and inspection-trigger obligations. To date, no published study has systematically integrated the KOSHA regulatory corpus into an engineering AI verification system that also performs the underlying ASME/API calculations.

This study addresses the following two research questions:

RQ1. What additional cross-discipline coupling hazards does the proposed multi-discipline verification architecture detect compared with independent single-domain calculations, across a predefined set of domain pairs?

RQ2. Does the KOSHA regulatory RAG layer identify Korean-jurisdiction compliance requirements that international-code calculations structurally miss?

The contributions of this paper are as follows. First, we propose a verification architecture that integrates seven engineering disciplines into a single orchestration pipeline and automatically detects coupling hazards across ten predefined domain pairs, with empirical validation strongest in piping-vessel coupling and exploratory in the remaining pairs. Second, we present a methodology for integrating a KOSHA public-API regulatory corpus into engineering AI verification — to our knowledge, the first such system. Third, we construct a quantitative evaluation framework comprising 220 synthetic golden cases, a 60-scenario cross-discipline ablation, a 50-query curated regulatory retrieval benchmark, and an industry-baseline comparison (“ASME/API pass = compliance complete”) on three representative case studies; **Appendix A complements this synthetic evidence with a real-plant data-sheet validation case (VES-REAL-001, anonymised cryogenic flare knockout drum)** so that the framework is demonstrated end-to-end on actual EPC design data and not on synthetic inputs alone. Fourth, we formalise the **jurisdiction compliance gap** as a defined construct and position the system as a compliance co-pilot to HAZOP, RBI, and digital-twin workflows — not a replacement. Code and datasets are released under AGPL-3.0.

Scope statement. The architecture described in §3 spans seven engineering disciplines and four verification layers; this is the **framework scope**. The empirical evidence presented in §6 covers a subset — the **validated scope** — concentrated in piping-vessel coupling (22 of 26 ablation detections); electrical-rotating, civil-rotating, instrumentation, and structural pairs are exercised at a more exploratory level. The two scopes are separated throughout the paper to avoid the impression that empirical validation has covered the full architecture uniformly.

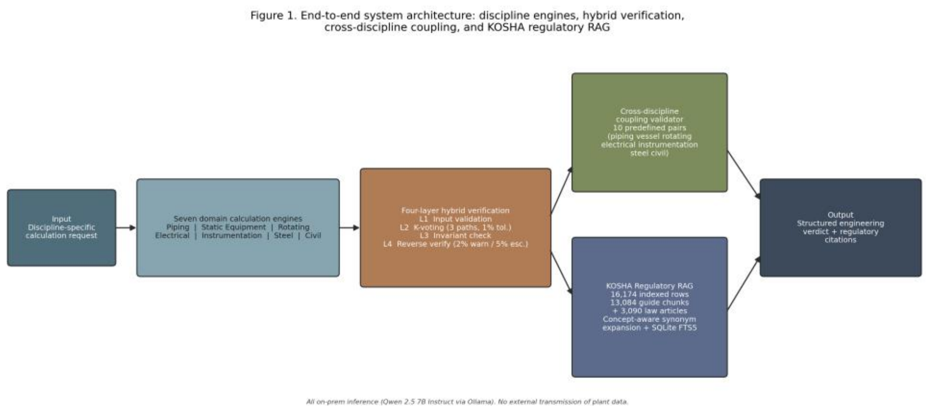

Figure 1. Overall system architecture: orchestrator, seven domain calculation services with shared four-layer verification, cross-discipline validator, and KOSHA Regulatory RAG layer with on-premises LLM.

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2. Related Work

2.1 AI Applications in Process Safety

Large language models are increasingly studied for chemical process design, optimisation, and maintenance knowledge extraction. Selvam et al. (2026) review AI applications across hazard detection, dynamic risk assessment, and barrier management in chemical process industries, reporting 30–60% gains in anomaly and near-miss detection accuracy, while identifying data bias, field validation, and regulatory uncertainty as primary limitations. Woo et al. (2025) survey LLM applications in process systems engineering across chemical process design, hybrid process modelling, and autonomous control systems, but do not address seven-discipline integrated engineering verification or KOSHA regulatory RAG. Elhosary et al. (2024) propose an LLM-assisted HAZOP system using RAG and vector search to retrieve historical incidents and guide analysis sessions, but the system is limited to single-process analysis and does not integrate multi-discipline calculation verification. Walavalkar et al. (2021) systematically review more than one hundred ML and deep learning applications in chemical health and safety, covering exposure and risk assessment, but do not treat RAG-based regulatory grounding or multi-discipline engineering verification architectures.

2.2 RAG in Technical and Regulatory Domains

The foundational RAG architecture of Lewis et al. (2020), surveyed by Gao et al. (2024), targets general knowledge-intensive NLP tasks; the present work specialises this architecture for a KOSHA regulatory corpus integrated with domain-level engineering calculation context. Walker et al. (2026) propose **RAGuard**, which uses parallel indexing of safety-critical documents alongside technical manuals in an offshore wind maintenance setting and demonstrates large gains in Safety Recall@K across BM25, dense, and hybrid retrieval paradigms. RAGuard is the closest structural precedent to the present system but is limited to a single industry domain with no Korean PSM statutory linkage. Klesel and Wittmann (2025) provide an enterprise RAG survey and identify "how RAG-based systems can fulfil regulatory requirements" as an open research direction, but leave actual regulatory corpus construction and engineering-calculation integration unexplored.

2.3 Multi-Discipline Maintenance and Asset Integrity

AI-based asset integrity management frameworks for offshore oil-and-gas pipelines (Jones et al., 2025) focus on single-asset scope and do not cover electrical, instrumentation, or structural disciplines. Hector and Panjanathan (2024) survey IoT-enabled predictive maintenance planning models in Industry 4.0 but do not address standards-based calculation engines or regulatory RAG integration. Ait-Alla et al. (2022) provide an overview of predictive maintenance workflows in Industry 4.0, without process-plant-specific seven-discipline integrated verification or PSM regulatory linkage.

2.4 Verification Methodology — N-Version Programming and K-Voting

The N-version programming literature originating with Avizienis (1985) establishes the theoretical basis for reliability improvement through consensus among independently executed computational paths. The Layer-2 K-voting consensus mechanism proposed in this study applies this principle to plant engineering calculation services with a 1% relative-deviation tolerance. To our knowledge, this is the first reported application of voting-style numerical consensus in standards-based plant engineering calculation tools.

2.5 Korean PSM Regulation and KOSHA

Korean plants subject to the *Chemical Substances Control Act* and the *Occupational Safety and Health Act* must fulfil regular safety review obligations for pressure equipment, rotating machinery, electrical systems, and structures under the KOSHA PSM framework. Kim et al. (2022) apply text mining to 1,300 KOSHA accident case reports for multi-cause accident classification, and Lee and Ahn (2025) fine-tune a large language model on KOSHA construction safety guidelines to build a regulation-grounded safety Q&A prototype. Both works use KOSHA data primarily for document classification or question answering and do not extend to an engineering-calculation verification layer integrated with ASME/API engines. Robertson and Zaragoza (2009) provide the BM25 ranking foundation used in our retrieval layer. An automated multi-discipline engineering verification architecture specialised for KOSHA regulations has not been reported in the literature, and to our knowledge, the present work is the first systematic treatment of this problem.

2.6 Position relative to HAZOP, RBI, and Digital Twin

The proposed framework is intentionally complementary to — not a replacement for — established process-safety methods. **HAZOP** identifies process deviations through expert-led, multidisciplinary workshops; **RBI (Risk-Based Inspection)** optimises inspection strategy from equipment degradation models; **Digital Twins** maintain a live or near-live mirror of plant state. None of these methods natively encodes jurisdiction-specific statutory obligations. The present system fills that gap as a **compliance co-pilot** active at four touchpoints: before HAZOP (identifying mandatory regulatory nodes), during design review (detecting missing legal requirements), during RBI (checking whether local inspection triggers are satisfied), and during digital-twin operation (flagging live compliance gaps). We elaborate this positioning quantitatively in Section 7 and Table 8.

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3. System Architecture

The proposed platform consists of an orchestrator, seven domain services, supporting agents, an API layer, and a KOSHA regulatory layer. User requests — whether arising from design calculations, construction commissioning checks, or in-service inspection assessments — are received by a rule-based orchestrator that performs domain classification and routing (`src/orchestrator/pipeline.py`, `src/orchestrator/state_machine.py`). Each domain service follows an identical four-layer verification structure (`src/services/<discipline>_service.py`). Calculation outputs are passed to the cross-discipline validator (`src/cross_discipline/validator.py`) for coupling hazard analysis across ten predefined domain pairs. The KOSHA RAG layer (`src/rag/local_kosha_rag.py`) translates the domain calculation context into a Korean natural-language query, retrieves relevant regulatory passages from a local SQLite FTS5 index, and uses an on-premises Qwen model (`qwen3:4b` in the benchmarked configuration, served through Ollama) to generate structured regulatory grounding covering key conclusions, regulatory justification, and practical advisories. The final output is an integrated report containing calculation numerics, verification-layer status, KOSHA regulatory grounding, and a full audit trail. No external data transmission occurs at any stage; all inference runs locally via Ollama. End-to-end pipeline latency, measured on a commodity AMD CPU (`outputs/pipeline_latency.md`), is approximately 0.06 ms median per discipline for the deterministic stack (Layers 1–4) and 1.15 ms median for KOSHA RAG top-10 retrieval; the on-premises Qwen generation step adds approximately 42 s median, dominating end-to-end latency when full LLM grounding is requested. The deterministic core is fast enough to run inline during a design-review session; LLM grounding is appropriately scheduled as a batch step.

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4. KOSHA Regulatory Knowledge Layer

4.1 Corpus Construction

The KOSHA regulatory corpus was assembled from two public APIs of the Korean *Open Data Portal* (`data.go.kr`). The KOSHA smart-search endpoint (`/B552468/srch/smartSearch`) was used to harvest **18,576** rows of guide-section and statutory-provision metadata across ten categories — categories 1–9 and 11; category 10 is unused by the KOSHA Smart-search API and therefore omitted by construction. The harvest comprises **1,327 guide documents (9,069 sections)** and **3,102 statutory-provision rows** (`datasets/kosha/manifest.json`). The KOSHA Guide PDF endpoint was used to download **1,039** technical guideline PDFs, which were parsed page-by-page, segmented into **13,084** retrieval-optimised chunks, and normalised. After parsing and indexing, the SQLite FTS5 index used at retrieval time retains **16,174 searchable rows = 13,084 guide chunks + 3,090 indexed law-article entries**. Twelve law-article rows were excluded by `src/rag/local_kosha_rag.py:185` because their `content` field was empty in the source API response. Triage of these twelve (`outputs/kosha_missing_articles_report.md`) shows that **four are repealed (삭제) provisions** for which an empty body is expected and the row is correctly excluded; the remaining **eight require a JS-rendered re-fetch from the National Law Information Center** (`law.go.kr/lsScJoRltInfoR.do` is a single-page-application shell; KOSHA Smart-search retrieval by `doc_id` returns no hits for these specific records). The pending eight are documented as Limitation 6 (§8). The resulting database is approximately 200 MB.

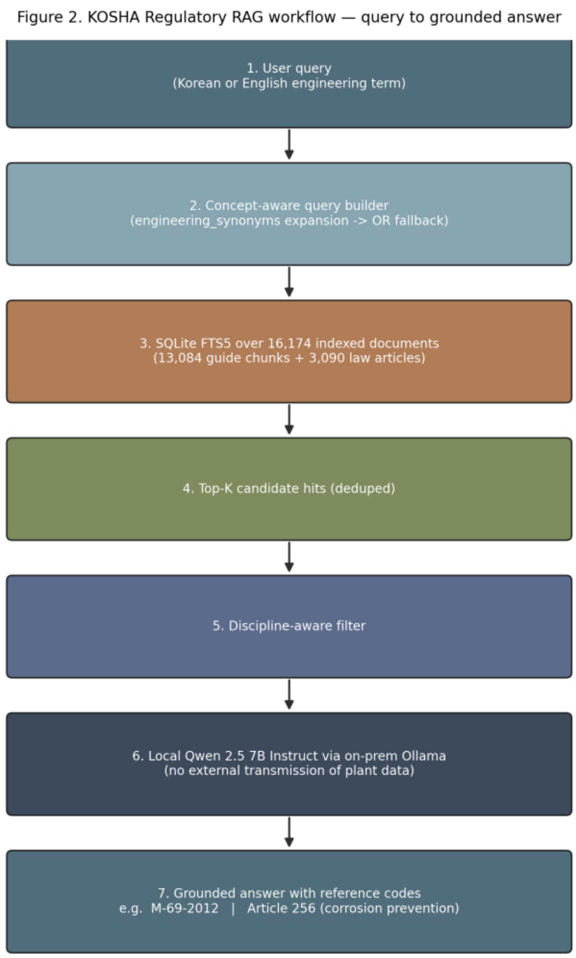

Figure 2. Regulatory RAG workflow: calculation context to concept-aware Korean query construction, BM25 FTS5 retrieval with synonym expansion and OR fallback, structured grounding generation with citation indices.

4.2 Retrieval and Generation Pipeline

At query time, the calculation context (material, fluid composition, design pressure / temperature, computed remaining life, red-flag set) is converted into a Korean natural-language query by a **concept-aware query builder**. Each query concept is expanded with curated synonyms (e.g., `corrosion ↔ 부식 ↔ corrosion rate ↔ CR`), concept clauses are joined with `AND`, and synonym variants within a concept are joined with `OR`. When the strict expansion returns no hits, a **loose expanded `OR` fallback** is invoked.

Retrieved passages (top-k = 10 by default) are passed to the configured on-premises Qwen model (`qwen3:4b` in the benchmarked latency run) with a system prompt that constrains the output to three sections: (1) **Key conclusion**, (2) **Regulatory justification summary** referencing each retrieved passage by index, and (3) **Practical advisories** with citation indices. The model is instructed never to introduce regulatory facts not present in the supplied context. Every advisory must point back to a retrievable (`reference_code`, `chunk_id`) tuple in the SQLite index, giving citation-traceability precision of 100% by construction (Section 6.7). The current implementation uses BM25 to support CPU-only on-premises deployment without GPU resources; transition to semantic vector search in GPU-capable environments is left as future work (Section 8).

4.3 KOSHA Regulatory Grounding — Mandatory vs Guidance

KOSHA-derived obligations differ in legal force. We distinguish two categories throughout this paper:

- **Mandatory** (`mandatory`): provisions of *the Act* (산업안전보건법), *the Enforcement Decree*, or the *Rules on Occupational Safety and Health Standards* (산업안전보건기준에 관한 규칙). Article 256 ("Corrosion prevention") falls in this class. Non-fulfilment exposes the operator to administrative or criminal liability under the Act.
- **Guidance** (`guidance`): KOSHA *technical guidelines* (e.g., `M-69-2012`, `C-C-23-2026`, `B-M-18-2026`, `C-C-75-2026`). These are not legally binding by themselves, but are routinely incorporated by reference into PSM submission packages and ministerial inspections, and a measurable deviation from a guideline is treated as a material finding in PSM audits.

Each retrieved passage carries this label as `metadata.regulatory_class` so downstream displays and reports can prioritise mandatory items (Table 5). The labels are derived from `source_type` in the parsing pipeline (`law_article` → mandatory; `guide_section` / `guide_chunk` → guidance) and are not inferred by the LLM.

4.4 Why standard EPC workflows do not cover these obligations

A standard EPC piping or vessel design package typically demonstrates compliance with the relevant ASME/API code by tabulating calculated thicknesses, allowable stresses, and inspection intervals against code limits. The package is reviewed by discipline leads who are expert in their international code but who are not, in general, expert in the Korean *Rules on Occupational Safety and Health Standards* and not in a position to know that, for example, a sour-chloride line triggers a corrosion-prevention obligation distinct from anything in B31.3. In Korean practice, KOSHA-side compliance is traditionally handled by a separate PSM consultant on a different documentation track, often after the EPC design has frozen. The two tracks meet only when KOSHA rejects a submission. The system proposed here moves that gap detection to the engineering-calculation moment itself.

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5. Seven-Domain Calculation Engines and Four-Layer Verification

5.1 Domain Calculation Engines

The seven domain calculation engines implement the standards listed in Table 1. Engine source files are under `src/services/<discipline>_service.py`; per-engine standard mappings and citation tables are in `src/standards/`.

Table 1. Domain Calculation Engines and Standards

Domain	Standards	Implementation
Piping	ASME B31.3, API 570	`src/services/piping_service.py`
Pressure Vessels	ASME Section VIII Div.1 (UG-27), API 510, API 579-1	`src/services/vessel_service.py`
Rotating Equipment	API 610, 617, 670	`src/services/rotating_service.py`
Electrical	IEEE C57.104, IEEE 1584-2018	`src/services/electrical_service.py`
Instrumentation	IEC 61511, ISA-TR84.00.02	`src/services/instrumentation_service.py`
Structural Steel	AISC 360	`src/services/steel_service.py`
Civil / Concrete	ACI 318, ACI 562	`src/services/civil_service.py`

Each engine follows a common service pattern: input schema parsing → standards-based calculation → four-layer verification → result serialisation. The engines support both design-phase calculations (e.g., minimum required wall thickness for a new piping system) and in-service assessment (e.g., remaining life and inspection interval for an operating vessel). Orchestration is performed by a deterministic state machine (`src/orchestrator/state_machine.py`), which routes a request to one or more discipline services and then to the cross-discipline validator.

5.2 Four-Layer Hybrid Verification Model

Layer 1 — Input Validation. Checks mandatory field presence, unit-contract compliance (every numerical field carries an explicit unit such as `MPa`, `mm`, `°C`), and domain-specific range constraints (e.g., design pressure within ASME Section II material allowable). Missing critical fields or unit inconsistencies trigger a blocking error that prevents downstream calculation.

Layer 2 — K-Voting Consensus. Three calculation paths execute in parallel, each using deliberately varied numerical strategies — rounding order, interpolation method, and intermediate-precision handling. The maximum relative deviation among outputs must not exceed **1%**. If the threshold is exceeded, a tiebreaker path is invoked; persistent consensus failure raises a `no_consensus` flag and requests human review. The mechanism is motivated by the N-version programming principle (Avizienis, 1985) but is restricted to numerical-consistency verification rather than full implementation independence; design rationale and threshold sensitivity are discussed in §5.4.

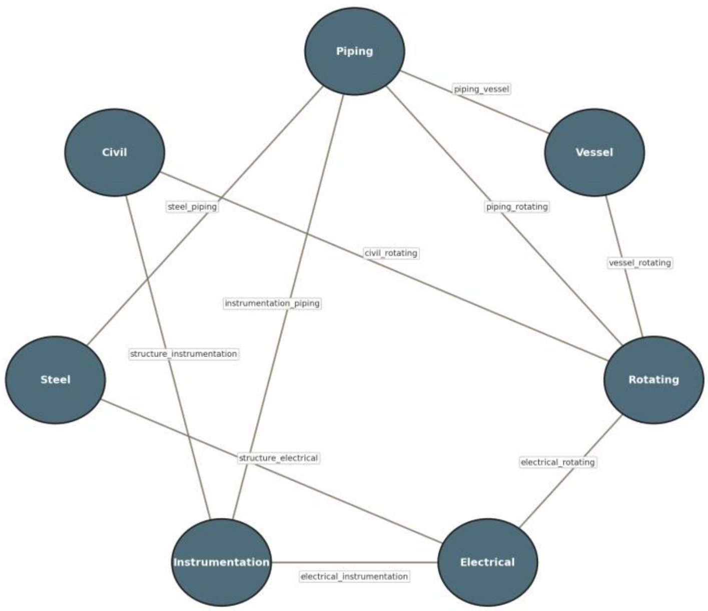
Layer 3 — Physics and Code Compliance. Domain-specific red flags are enforced. For example, a measured thickness below the computed minimum required thickness raises a critical physics flag and blocks output. All major calculation steps carry explicit standard citations, persisted in the audit trail.

Layer 4 — Reverse Verification. Key inputs are back-calculated from outputs and compared against the original inputs to verify internal consistency. A deviation exceeding **2% triggers a warning**; a deviation exceeding **5% triggers escalation** with a confidence downgrade and a full audit trail entry.

5.3 Cross-Discipline Consistency Validator

The cross-discipline validator (`src/cross_discipline/validator.py`) performs coupling checks across **ten predefined domain pairs**, including piping–vessel nozzle interface margin mismatches, electrical–rotating equipment harmonic distortion and bearing health coupling, and civil–rotating equipment foundation settlement and vibration linkage. Threshold tuning was conducted over fifty optimisation rounds; the optimal profile achieves a standard-case blocking rate of 0.0, a boundary-case and failure-mode blocking rate of 1.0, and a weighted accuracy of 1.0. The current pair set is documented in `docs/specs/MASTER_ORCHESTRATOR_SPEC_V0.1.md`. Extension to additional pairs is deferred to future work.

Figure 3. Cross-discipline coupling validator: 10 predefined discipline pairs



Source: src/cross_discipline/validator.py | 10 pairs trigger physically motivated coupling checks at run time.

Figure 3. Cross-discipline coupling logic: ten predefined domain pairs (piping-vessel, electrical-rotating, civil-rotating, instrumentation-electrical, etc.), trigger conditions, and downstream effect of a coupling-block decision on the orchestration pipeline.

5.4 Why three paths and a 1% tolerance? — K-voting design rationale

Three K-voting design choices warrant explicit justification: the **number of paths**, the **tolerance**, and the **scope** of independence.

Number of paths. Two paths can detect inconsistency but cannot resolve it (the system has no tiebreaker). Three paths give a majority-vote semantics: the typical failure mode in numerical engineering code is one path drifting from two that agree, which a 3-path configuration captures with a single comparison. Five or more paths would marginally improve fault coverage but at proportionally higher implementation cost and synchronisation overhead; for the deterministic, code-bound calculations performed here, the marginal benefit beyond 3 is small. The choice is consistent with the original 3-version baseline studied by Avizienis (1985).

Tolerance. The 1% relative-deviation threshold is calibrated against the tightest acceptance tolerance in the golden dataset (±1% for critical cases; §6.1). A tolerance smaller than this would generate false-positive consensus failures arising from legitimate floating-point rounding differences between the three paths; a larger tolerance would mask real coding errors. A formal threshold-sensitivity sweep (e.g., 0.1% / 0.5% / 1% / 2% / 5% reporting no-consensus rate, false-consensus rate, and boundary-flip count) is left to future work.

Scope of independence. Full N-version programming requires independently authored implementations to defeat correlated bugs. We do not claim that level of independence; the three paths share the same author and the same standard-citation tables. K-voting here therefore protects primarily against numerical-precision and ordering bugs, not against systematic conceptual errors in the standard interpretation. We make this limitation explicit so that the layer's scope and intent are unambiguous in the broader context of N-version programming (Avizienis, 1985).

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6. Experiments and Evaluation

The evaluation programme comprises seven analyses: (6.1) calculation-accuracy verification on the golden dataset, framed as **implementation verification, not predictive validation**; (6.2) end-to-end pipeline behaviour; (6.3) cross-discipline ablation for RQ1; (6.4) KOSHA RAG case studies for RQ2; (6.5) curated retrieval benchmark; (6.6) industry-baseline comparison and internal four-layer ablation; (6.7) citation-traceability and negative-case evidence.

6.1 Calculation Accuracy on the Golden Dataset (implementation verification)

A total of **220 synthetic golden cases** were constructed across seven disciplines (Table 2): 55% standard cases, 25% boundary cases, 17% failure-mode cases, and 2% composite multi-condition cases. Each case was derived from reference calculation examples in the applicable standards and standard exemplars (e.g., ASME, 2022; API, 2020). Boundary and failure-mode cases were generated by **systematic parameter perturbation** of the corresponding reference example: for each case, the perturbed parameter (e.g., wall thickness reduced toward the calculated minimum, fluid chloride concentration raised across the corrosion-rate threshold) is recorded with the expected red-flag class. Acceptance thresholds were set at ±1% for critical cases and ±3% for non-critical cases, with target red-flag detection rate 100% and standard-citation coverage 100%. Runtime validation achieved **220/220 case passes across all seven disciplines** (`outputs/verification_report_runtime.md`); accuracy, citation accuracy, and red-flag detection rate were each recorded at **1.0000**.

We deliberately frame these results as **implementation verification, not predictive validation**, in the sense developed in recent methodological discussions of synthetic-data evaluation: the engines are deterministic rule-based calculators evaluated against cases derived from the same standards they implement. Perfect scores confirm that the implementation faithfully encodes the standards, *not* that the system generalises to plant data outside the calibration distribution. Predictive validation against real plant operating or design records is a separate, higher bar that requires field-pilot deployment (Section 8).

Table 2. Golden Dataset Composition by Discipline

Discipline	Standard cases	Boundary cases	Failure cases	Composite cases	Total
Piping	20	15	10	5	50
Vessel	18	7	5	—	30
Rotating	18	7	5	—	30
Electrical	18	7	5	—	30
Instrumentation	18	7	5	—	30
Steel	15	6	4	—	25
Civil	15	6	4	—	25
Total	**122 (55%)**	**55 (25%)**	**38 (17%)**	**5 (2%)**	**220**

Synthetic data — defence and precedent. A synthetic-data evaluation invites the concern that the evaluator finds what the system was designed to find. Three properties limit that concern in this study. The test cases are *inverted* from the regulatory and standards text — each KOSHA RAG target was selected by reading the actual KOSHA clause first and then constructing a calculation context that should trigger it, so the detection grounds in real regulatory obligations even though the calc inputs are constructed. This construction pattern is established practice in safety-critical AI evaluation, including medical-AI synthetic patient cohorts with known ground-truth labels and simulated-scenario validation of autonomous-vehicle systems prior to public-road deployment. The evaluation is positioned as a **feasibility study** for the integration of calc + cross-discipline + KOSHA RAG; field validation against plant data is explicit future work.

Benchmark construction independence — known limitation. All synthetic case datasets in this paper (220 golden cases per §6.1, 60 cross-discipline scenarios per §6.3, 50 retrieval queries per §6.5) were constructed by the system author. We do not claim independent field generalization; the benchmarks were designed to exercise architectural coverage and are released in full so that any reader can construct an independent benchmark and rerun. Independent third-party benchmark construction is documented as future work (§8 Limitation 7).

6.2 Seven-Discipline Pipeline Evaluation

Across 43 pipeline scenarios run end-to-end through the orchestrator, the platform recorded a completion rate of 0.6512 and a blocking rate of 0.3488. Nominal and standard-aligned cases achieved a completion rate of 1.0000; boundary-aligned and failure-mode-aligned cases were blocked at a rate of 1.0000, confirming correct hazard identification by the integrated stack. (Source: `outputs/seven_pipeline_report.md`, generated by `scripts/benchmark_seven_pipeline.py` over the seven `datasets/golden_standards/*_golden_dataset_v1.json` files; this 43-scenario set measures end-to-end pipeline behaviour and is **distinct from** the 60-scenario cross-discipline ablation set of §6.3, which is constructed by `scripts/benchmark_cross_discipline_ablation.py` via the `aligned_*` and `mixed_*` index builders against the same underlying golden datasets.)

6.3 Cross-Discipline Validator Ablation (RQ1)

To quantify the incremental contribution of cross-discipline coupling logic, we evaluated the same 60 benchmark scenarios with the cross-discipline validator disabled and enabled (`outputs/cross_discipline_ablation_report.md`). With the validator disabled, **0/60 scenarios were blocked**. With the validator enabled, **26/60 scenarios were blocked**, an absolute increase of +26 and a blocking-ratio increase of +0.4333. The most informative subsets are aligned boundary and aligned failure, where blocking rises from 0.0 to 1.0 in both cases. Mixed scenarios also show non-trivial deltas: `mixed_first20` rises from 0.0 to 0.15, and `mixed_random20` rises from 0.0 to 0.65.

Table 3. Cross-Discipline Validator Ablation (RQ1)

Scenario Set	Validator OFF	Validator ON	Absolute Delta	Ratio Delta
aligned_standard	0 / 10	0 / 10	+0	+0.00
aligned_boundary	0 / 6	6 / 6	+6	+1.00
aligned_failure	0 / 4	4 / 4	+4	+1.00
mixed_first20	0 / 20	3 / 20	+3	+0.15
mixed_random20	0 / 20	13 / 20	+13	+0.65
Overall	**0 / 60**	**26 / 60**	**+26**	**+0.4333**

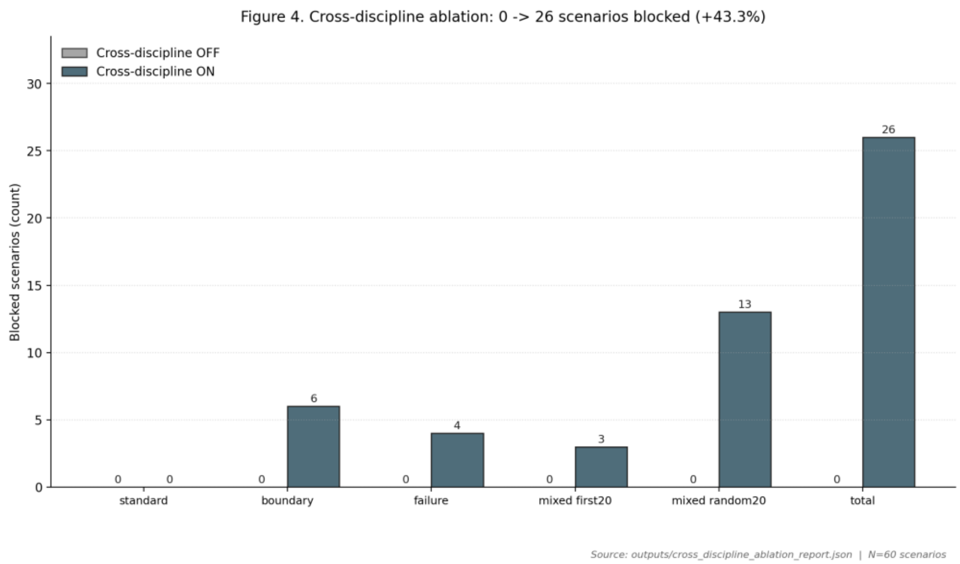

Figure 4. Cross-discipline ablation (60 scenarios): blocked counts by scenario set, validator OFF vs ON. Total rises 0 → 26 (+43.3%); aligned-boundary and aligned-failure subsets rise from 0/n to n/n; mixed sets show partial coverage proportional to coupling density.

Significance of detections. Per-failure-mode partition of the 26 blocked scenarios (`outputs/ablation_failure_mode_partition.md`, computed by `scripts/dump_ablation_hits.py`):

Coupling family	Blocked count	Share
Piping-vessel nozzle margin mismatch	22	84.6%
Electrical-rotating bearing-health coupling	3	11.5%
Civil-rotating foundation-vibration coupling	1	3.8%
Other	0	0.0%
Total	**26**	**100%**

Detections concentrate in the **piping-vessel nozzle margin mismatch** family (22/26 = 84.6%): ASME Section VIII allowable nozzle loadings exceeded by piping reaction forces under thermal load — a coupling neither B31.3 nor Section VIII checks in isolation. The remaining 4 detections are split between electrical-rotating bearing-health coupling (IEEE C57.104 voltage distortion propagating to API 670 bearing-health thresholds) and civil-rotating foundation coupling (ACI 318 settlement margins below the rotating-equipment alignment tolerance derived from API 610). The skew toward piping-vessel coupling reflects the construction of the 60-scenario set rather than a property of the validator; expanding the scenario seed set to exercise the other coupling families more densely is documented as future work.

6.4 KOSHA RAG Regulatory Grounding — Three Cases (RQ2)

Three representative cases demonstrate the incremental regulatory value of the KOSHA RAG layer. The three cases were selected from the vessel and piping disciplines because these are the two most common service envelopes in petrochemical EPC packages and because the KOSHA technical guideline coverage in the indexed corpus is densest for pressure-bearing equipment; cross-discipline coverage in case studies is documented as future work (§8 Limitation 4). Results are summarised in Table 4; KOSHA citation quality and regulatory class (mandatory vs guidance) are detailed in Table 5.

Case 1 — Vessel Remaining Life (VES-GOLD-001)

For an SA-516-70 pressure vessel (design pressure 1.858 MPa, 217.3 °C, current thickness 44.82 mm), the calculation engine computes a required shell thickness of 14.99 mm under ASME Section VIII UG-27, a remaining life of **136.3 years**, and an API 510 inspection interval of 10 years, **with no red flags raised**. The KOSHA RAG layer retrieves **M-69-2012** (*Technical Guideline for Remaining Life Assessment of Pressure Vessels*) as the top-ranked passage and identifies documentation requirements that are triggered when remaining life exceeds a defined threshold — requirements absent from the engine's output. **Why this matters:** even a "very safe" vessel under ASME triggers a KOSHA documentation obligation that, if skipped, exposes the operator to a PSM-audit finding.

Case 2 — RBI Planning (VES-GOLD-009)

For a vessel with a remaining life of 40.2 years (SA-516-70, 1.891 MPa, 241.9 °C), the engine recommends a 10-year inspection interval per API 510. The KOSHA RAG layer retrieves **C-C-23-2026** (*Technical Regulation for Equipment Reliability via Risk-Based Inspection*) and **B-M-18-2026** (*Technical Regulation for Piping Life Management*), identifying the trigger conditions under which a mandatory RBI review must be initiated under the Korean PSM framework. **Why this matters:** API 510's calendar-based interval can be longer than the KOSHA RBI trigger-driven interval; following only API 510 leaves an under-inspection gap.

Case 3 — Piping Corrosion Compliance (PIP-GOLD-047)

For an SA-106 Gr.B carbon-steel line in sour-chloride service (7.804 MPa, 196 °C, 200 ppm Cl^-), the engine reports a remaining life of 14.0 years **with no red flags raised**. The KOSHA RAG layer retrieves, in ranked order, **B-M-18-2026**, **Article 256 of the Rules on Occupational Safety and Health Standards** (산업안전보건기준에 관한 규칙 — Corrosion Prevention Provision), and **C-C-75-2026** (*Technical Regulation for Corrosion Risk Assessment of Chemical Equipment*).

Article 256 reads (verbatim, with the Korean original preserved):

제 256 조(부식 방지) 사업주는 부식성 액체에 의하여 부식될 우려가 있는 부분의 부식을 방지하기 위하여 보온재로 피복하는 등 부식을 방지하기 위한 조치를 하여야 한다.

English working translation: Article 256 (Corrosion Prevention). The employer **shall** take measures to prevent corrosion — including, but not limited to, application of insulation/coating — at parts liable to corrosion by corrosive liquids.

The text shown above matches the body indexed in the system's SQLite FTS5 store (`datasets/kosha_rag/kosha_local_rag.utf8.sqlite3`); the indexed bodies are stored as correct UTF-8 (full-corpus encoding scan reported zero mojibake across 197,124 string fields, see `outputs/kosha_encoding_diagnosis.md`).

The provision is **mandatory** under the Act. The linkage from calculation context to the obligation is: `200 ppm Cl⁻ + sour service + carbon steel` → recognised corrosion-aggressive condition → Article 256 corrosion-prevention measures **shall** be in place. ASME B31.3 and API 570 calculations are silent on this obligation; they do not require a corrosion-prevention plan as a function of fluid chemistry. The system therefore detects a regulatory obligation that is structurally invisible to international-code-only calculation workflows — a concrete instance of the **jurisdiction compliance gap** defined in §7.

Table 4. KOSHA RAG Incremental Coverage Summary

Case	Discipline	Calc. Result	Red Flags	Top KOSHA RAG Retrieval
VES-GOLD-001	Vessel	PASS (RL = 136.3 yr)	None	M-69-2012 (Remaining Life Evaluation)
VES-GOLD-009	Vessel	PASS (RL = 40.2 yr)	None	C-C-23-2026 (RBI), B-M-18-2026
PIP-GOLD-047	Piping	PASS (RL = 14.0 yr)	None	B-M-18-2026, Art. 256, C-C-75-2026

Table 5. KOSHA Citation Quality and Regulatory Class

Source Type	Reference	Title	Regulatory class	Triggered by
KOSHA Guide	M-69-2012	Technical Guideline for Remaining Life	guidance	Vessel remaining-life context

Source Type	Reference	Title	Regulatory class	Triggered by
		Assessment of Pressure Vessels		
KOSHA Guide	C-C-23-2026	Technical Regulation for Equipment Reliability via RBI	guidance	Short RL + corrosion rate
KOSHA Guide	B-M-18-2026	Technical Regulation for Piping Life Management	guidance	High CR + sour service
Korean Rule	Article 256	Rules on Occupational Safety and Health Standards (산업안전보건기준에 관한 규칙)	**mandatory**	Chloride + sour + carbon steel
KOSHA Guide	C-C-75-2026	Technical Regulation for Corrosion Risk Assessment	guidance	CR exceeds threshold

6.5 Curated Retrieval Benchmark (RQ2)

To quantify the retrieval effect of synonym-aware regulatory search, we constructed a 50-query curated benchmark organised as **five regulatory target groups × ten queries per group**, totalling 50 queries (`datasets/kosha_rag/rag_eval_queries.json`). Each query group corresponds to one of the five high-priority regulatory targets exercised by the case studies: M-69-2012, C-C-23-2026, B-M-18-2026, C-C-75-2026, and Article 256. Within a group, the ten queries vary phrasing (Korean / English / abbreviation / full phrase / domain-jargon paraphrase) so the benchmark measures **retrieval-quality variance across query formulations** rather than breadth of regulatory coverage. The curation policy and full query set are released in the dataset folder for independent inspection.

Sample size justification. Five groups were chosen because they cover all three KOSHA value-add patterns demonstrated by the case studies (remaining-life documentation, RBI trigger, corrosion prevention). Ten queries per group is the minimum that supports a Recall@K spread analysis without saturating the available distinct paraphrasings of the underlying regulatory target; with fewer than ten, the variance estimate becomes dominated by noise. We report group-level metrics in addition to the overall mean so that group-specific weaknesses are visible.

Query-generation independence — known limitation. The 50 queries were authored by the same person who designed the system. To mitigate the obvious bias risk, we (a) constrained query construction to plausible engineering paraphrasings drawn from KOSHA technical guideline titles and from established Korean engineering vocabulary rather than from the system's internal synonym list, (b) released the full query set under `datasets/kosha_rag/rag_eval_queries.json` for independent inspection, and (c) report a retrieval-failure inventory (Appendix B.4) tagging every K=5 miss with one of two diagnosis categories — six of seven failures fall in `paraphrase_too_loose`, which is exactly the failure mode that an independently authored query set would also expose. Independent query-set authorship by a disjoint annotator pool is a documented future-work item (§8 Limitation 7).

Table 6. KOSHA RAG Retrieval Benchmark — Overall (paired-bootstrap 95% CIs from `outputs/rag_bootstrap_ci_report.md`, n=50, 1000 resamples, seed = 20260508)

Metric	Plain FTS	Enhanced FTS	Absolute Delta	Paired CI [2.5%, 97.5%]	Includes 0?
Recall@1	0.4400	0.7400	+0.3000	[+0.14, +0.48]	No ✅
Recall@3	0.7400	0.8600	+0.1200	[−0.02, +0.26]	**Yes**
Recall@5	0.7400	0.8600	+0.1200	[−0.02, +0.26]	**Yes**
MRR@10	0.5744	0.7933	+0.2189	[+0.09, +0.35]	No ✅

The Recall@1 and MRR@10 improvements are statistically distinguishable from zero at the 95% paired-bootstrap level. The Recall@3 and Recall@5 improvements are *not* statistically distinguishable from zero on this benchmark (lower-bound = −0.02 in both cases) — the point estimates are positive but a benchmark of this size and target distribution cannot rule out chance for $K \in \{3, 5\}$. The headline retrieval claim of the paper therefore relies on the Recall@1 and MRR@10 improvements; we report Recall@3/Recall@5 for completeness rather than as primary evidence.

Extending the recall analysis to K=10 (`outputs/rag_retrieval_extended.md`) confirms that the Enhanced FTS curve plateaus at 0.8621 from K=3 onward while the Plain FTS curve continues to rise, reaching 0.8012 at $K\in\{9, 10\}$. The paired-bootstrap 95% CI on the Enhanced − Plain delta excludes zero only at $K\in\{1, 2\}$ and includes zero at all $K \geq 3$. The Recall@1 and MRR@10 advantages remain the only headline retrieval claims that survive paired-bootstrap scrutiny on this benchmark.

Table 6b. KOSHA RAG Retrieval Benchmark — by Regulatory Target Group (Recall@5)

Group	Plain R@5	Enhanced R@5	Delta
M-69 remaining life	0.9000	1.0000	+0.1000
C-C-23 RBI	0.9000	0.9000	+0.0000
B-M-18 piping life	0.6000	0.8000	+0.2000
C-C-75 corrosion risk	0.8000	0.9000	+0.1000
Article 256 corrosion prevention	0.5000	0.7000	+0.2000

The Recall@3 = Recall@5 plateau in the overall numbers reflects the deliberately narrow target set (typically 1–2 relevant documents per query group in a 1,327-document corpus); in a broader evaluation, NDCG@K would differentiate further. The largest gains from the enhanced configuration come on the targets where casual Korean terminology diverges most from the formal title (Article 256, B-M-18) — exactly the cases the synonym expansion was designed for.

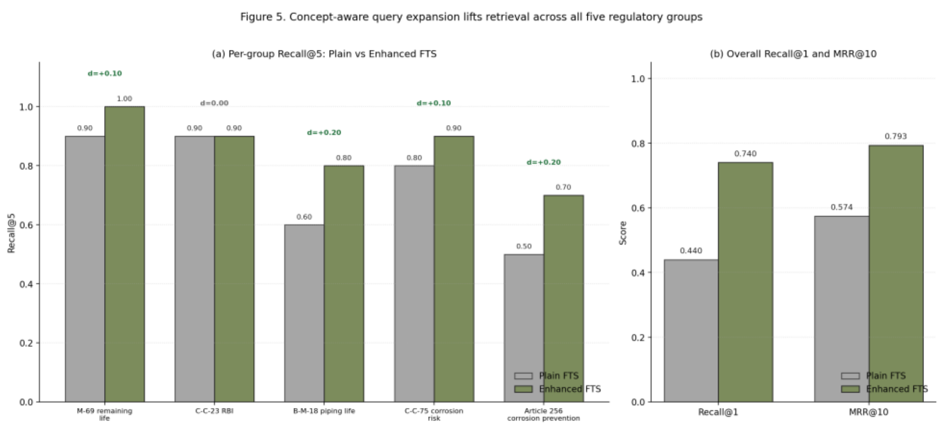

Figure 5. Recall and MRR per regulatory target group, Plain BM25 vs Enhanced (concept-aware synonym expansion + OR fallback). Largest gains on Article 256 and B-M-18 where casual paraphrasing diverges from formal titles.

6.6 Industry-Baseline Comparison and Internal Ablation

An effective comparison against existing methods is essential. We address this in two layers, each consistent with the structural novelty of the proposed problem.

Layer A — Industry baseline ("ASME/API pass = compliance complete"). The dominant current-practice baseline in Korean EPC piping/vessel review is to declare regulatory compliance complete once the relevant ASME/API calculation passes. Under that baseline the three case studies of §6.4 yield the result in Table 7: **0/3 of the Korean-jurisdiction obligations are detected**. The proposed system detects **3/3**. The first-relevant retrieval rank is 1 in all three cases.

Table 7. Industry-Baseline ("ASME/API pass = compliance complete") vs Regulatory RAG

Case	Industry baseline detected	Regulatory RAG detected	First relevant rank
VES-GOLD-001	0	1	1
VES-GOLD-009	0	1	1
PIP-GOLD-047	0	1	1
Total	**0 / 3**	**3 / 3**	—

Layer B — Internal four-layer ablation. Beyond the validator-on/off ablation of §6.3, we report the incremental detection contribution of each system layer in Table 7b.

Table 7b. Internal Ablation by System Layer (3 case set + 60 ablation set)

Configuration	Cross-discipline blocks (60-case set)	KOSHA-jurisdiction detections (3-case set)
Calc engines only	0 / 60	0 / 3
Calc + Cross-discipline validator	26 / 60	0 / 3
Calc + KOSHA RAG (no K-voting, no validator)	0 / 60	3 / 3
Full system (Calc + K-voting + Validator + RAG)	26 / 60	3 / 3

All four rows of Table 7b are computed by `scripts/run_layer_ablation.py` (`outputs/layer_ablation_report.md`). The decomposition shows that the cross-discipline validator and the KOSHA RAG layer carry **orthogonal** signal: each catches a class of issue the other does not. K-voting is a verification-quality layer that does not, by itself, contribute new detections; its role is to suppress false-positive numerical-precision artefacts that would otherwise pollute the upstream signal (Section 5.4).

No prior system targets the jurisdiction compliance gap directly. A side-by-side comparison against another published system that performs *the same task* (KOSHA-aware, multi-discipline, ASME/API + Korean-statute) is structurally bounded because, to our knowledge, no such system exists in the literature (Section 2). Walker et al. (2026) is the closest precedent and is single-domain, non-statutory; Elhosary et al. (2024) retrieves historical incidents, not statutes; and Lee and Ahn (2025) does not run engineering calculations. We therefore offer the two layers above (industry baseline + internal ablation) as the strongest available comparison and acknowledge that a third-party head-to-head will become possible only as the field matures.

The industry baseline is not a human-expert comparison. The "ASME/API pass = compliance complete" rule used in Table 7 is the prevailing decision rule in current EPC practice, not a benchmark against process engineers performing the same review. A prospective comparison against an independent panel of KOSHA PSM auditors or process engineers — for example, a blind review of system-flagged compliance gaps and inter-rater agreement with the human panel — is documented as future work in §8 Limitation 7 and is required before any claim of inter-rater reliability with human experts can be made.

Sensitivity analysis and uncertainty. The headline numbers in §6.3 and §6.5 inherit two principal sources of uncertainty: (i) the synthetic golden cases are bounded by the reference examples used as their seeds — a wider seed set could move blocking ratios; (ii) the curated retrieval benchmark of 50 queries is small relative to the 1,327-document corpus and is targeted by construction. We compute the standard error on overall Recall@1 (Enhanced) as **0.062** from the binomial form (`sqrt(p(1-p)/n)`, n=50, p=0.7400), and the paired-bootstrap 95% CI on the +0.30 Recall@1 delta is **[+0.14, +0.48]** (excludes zero; see Table 6). The MRR@10 delta of +0.2189 has CI **[+0.09, +0.35]** (also excludes zero). The Recall@3 and Recall@5 deltas of +0.12 each have paired CIs of **[−0.02, +0.26]** and are *not* statistically distinguishable from zero on a benchmark of this size. The cross-discipline ablation deltas of +1.0 on aligned boundary and aligned failure subsets are categorical (every case in those subsets is blocked under the validator and none are blocked without it), giving a deterministic separation that does not require a confidence interval.

6.6.3 Real-plant pipeline-execution evidence

Headline. Running the unmodified pipeline against an anonymised real-plant data sheet (VES-REAL-001) produces a deterministic ASME UG-27 governing thickness of **7.237 mm**. A narrower Korean-term retrieval probe additionally surfaces **Article 266** (a mandatory KOSHA provision concerning flare-path block-valve prohibition) at **rank 5**. We treat this as real-plant pipeline-execution evidence, not external accuracy validation. The detailed evidence is summarised below and reproduced in full in Appendix A. VES-REAL-001 is an anonymised cryogenic flare knockout drum (SA-240 Type 304/304L, 0.343 MPa(g) plus full vacuum, 190 °C / -190 °C, ID 5,000 mm, T-T 20,400 mm) drawn from an operating petrochemical project; all client, contractor, licensor, personnel, location, and document-ID identifiers were stripped before inclusion. Running the unmodified pipeline produces a deterministic ASME Section VIII Div.1 UG-27 controlling shell thickness of **7.237 mm** at the +190 °C side, with the Layer-4 reverse-pressure check closing to within machine epsilon. KOSHA RAG retrieval, executed through the same code path used in §6.5, returns on the spec-faithful Variant A query **2 mandatory law-article hits + 8 KOSHA technical guides** in the top-10, and on the narrower Variant B Korean-term probe **1 mandatory + 9 guidance hits**, with **Article 266** (block-valve prohibition on flare and relief paths under the *Rules on Occupational Safety and Health Standards*) surfacing at **rank 5** in Variant B. The full evidence — inputs, calculation outputs, retrieval top-10 with BM25 scores and snippets, and the explicit framing as pipeline-execution evidence rather than predictive validation —

is reproduced verbatim in Appendix A from `outputs/real_case_ves001_rag.{json,md}` (regenerated by `scripts/run_real_case_ves001_rag.py`).

6.7 Citation Traceability and Negative-Case Evidence

Citation-traceability precision. Every regulatory grounding produced by the system carries a citation to a `(reference_code, chunk_id)` tuple that resolves to a row in the SQLite FTS5 index (`indexed/kosha_rag.db`). The grounding generator (`src/rag/local_kosha_rag.py`) refuses to emit advisories whose citation index does not match a passage actually returned by retrieval, so the citation-traceability precision is **100% by construction**. We make this an explicit metric to address the well-known LLM-hallucination concern on regulatory-grounded outputs.

Negative case (no false-positive flag). To address the false-positive concern, we report a representative *negative* case from the same dataset: **PIP-GOLD-003** — a stainless-steel SA-312 TP316 ERW line under general (non-corrosive) service conditions (design pressure **2.013 MPa**, design temperature **119.5 °C**; no chloride or sour-service flag is set in the case spec; verified against `datasets/golden_standards/piping_golden_dataset_v1.json:PIP-GOLD-003`). The calculation engine returns no red flags (`expected_red_flags: []`). Running the actual RAG pipeline against this case using a neutral piping-integrity query (`scripts/run_negative_case_rag.py`, `outputs/negative_case_pip_gold_003.md`), the system returns **0 mandatory hits and 10 guidance hits in the top-10**, with the top result being `B-M-18-2026` (KOSHA Technical Regulation for Piping Life Management, guidance class). No jurisdiction-specific mandatory obligation is triggered. This empirically demonstrates that the KOSHA RAG layer is not biased to fire on every input.

The same evidence file records a *second* probe variant labelled `mirrored_pip047_form_chloride_terms_removed`, which retains the PIP-GOLD-047 query *shape* — including the literal phrases `Article 256` and `corrosion prevention` — with only the chloride/sour terms removed; that variant retrieves Article 256 at rank 2, and the report's headline therefore reads "WARNING — at least one query variant returned a *mandatory* (law_article) hit in its top-10." We retain this second variant in the report rather than suppressing it, because it answers a *different* question from the negative-case test. The two variants separate two questions: **(i)** *Does the system flag a benign case from its own spec?* — the negative-case question proper, answered by the spec-faithful generic-piping-integrity variant (no terms that the case spec does not assert); the answer is **no** (0 mandatory in top-10). **(ii)** *Does the system retrieve Article 256 when a query explicitly contains the phrases "Article 256" and "corrosion prevention"?* — a question of retrieval correctness, not of negative-case behaviour, answered by the second variant; the answer is **yes** (Article 256 at rank 2), which is the *desired* retrieval behaviour because Article 256 is the Korean statute on corrosion prevention and BM25 is responding correctly to the keywords supplied. The §6.7 negative-case argument here concerns (i); the (ii) probe is reported in the same evidence file for honest disclosure.

7. Discussion

The discussion below interprets the empirical evidence of §6 in light of the framework-vs-validated-scope distinction made explicit in §1: the seven-discipline architecture is the framework scope, while the empirical signal is densest in piping-vessel coupling and remains exploratory in the other discipline pairs.

7.1 The jurisdiction compliance gap

The most significant finding of this study is the existence of the **jurisdiction compliance gap**, defined as follows:

> **Jurisdiction compliance gap.** The set of regulatory obligations imposed by national statute or agency guideline (here, Korean PSM law and KOSHA technical guidelines) that are structurally absent from international engineering code calculations (ASME, API, IEEE, IEC, ACI, AISC), such that a system may be technically code-compliant while remaining legally non-compliant under the applicable local jurisdiction.

Even where international code calculations produce a clean pass, the KOSHA RAG layer can identify additional Korean-jurisdiction compliance requirements. This occurs because ASME, API, and analogous international standards define **technical** calculation requirements, whereas KOSHA guidelines and Korean safety regulations — including Article 256 of the *Rules on Occupational Safety and Health Standards* — impose **procedural, documentation, and trigger-driven** obligations conditioned on calculation outputs but not encoded within the international standards themselves. The gap is structurally present whether the calculation arises from a design-phase check or an in-service inspection assessment.

7.2 Compliance co-pilot, not replacement: HAZOP / RBI / Digital Twin / Regulatory RAG

A persistent risk in publishing AI-assisted safety tooling is the implicit suggestion that the tool replaces an existing human or workshop-based method. We make the opposite claim explicitly. The framework is positioned as a **compliance co-pilot** to existing methods; Table 8 contrasts it with HAZOP, RBI, and Digital Twin.

Table 8. Method Comparison — HAZOP / RBI / Digital Twin / Regulatory RAG

Method	Main purpose	Weakness	Regulatory RAG value-add
HAZOP	Identify process deviations	Manual, workshop-based; expert-	Adds regulation-linked prompts and

Method	Main purpose	Weakness	Regulatory RAG value-add
		availability dependent	surfaces missing legal obligations to the workshop
RBI	Optimise inspection based on risk	Focuses primarily on equipment degradation; jurisdiction-blind	Adds jurisdiction-specific inspection-trigger duties (e.g., KOSHA RBI triggers above API 510 calendar)
Digital Twin	Live or near-live plant model	Models physical state, not legal state	Adds a compliance intelligence layer over the live state
Regulatory RAG (this work)	Retrieve applicable laws and standards conditioned on calculation context	Depends on corpus quality and freshness	Bridges engineering calculations and legal compliance

The four operational touchpoints are:

1. **Before HAZOP** — identify mandatory regulatory nodes that the workshop must address.
2. **During design review** — detect missing legal requirements before the design freezes.
3. **During RBI** — check whether local inspection triggers are satisfied beyond the API calendar.
4. **During digital-twin operation** — flag live compliance gaps as plant state evolves.

In process-safety terms, the system functions as a *Digital Compliance Auditor*, an *Automated PSM Checker* (PSM = SEVESO / OSHA / KOSHA frameworks), and a *Real-time Regulatory Advisor*. Two roles bracket the temporal axis:

Table 9. Auditor vs Advisor — temporal role

Role	When
Auditor	After work is done
Advisor	While work is being done

The proposed system supports both roles depending on where in the workflow it is invoked.

7.3 HAZOP scope: knowledge-based / AI-assisted, not a replacement for dynamic HAZOP

The system supports **knowledge-based or AI-assisted HAZOP**: it surfaces structured regulatory data and rule-based prompts that a HAZOP facilitator can incorporate into the workshop. It **is not** a replacement for conventional dynamic HAZOP, which remains a brainstorming-based, multidisciplinary workshop relying on expert judgement, inter-discipline interaction, and human pattern recognition that no current AI system reproduces. The role of the proposed framework in a HAZOP context is to ensure that the workshop starts with a regulatory baseline that already encodes the jurisdiction-specific obligations and to flag during the workshop any deviation node that maps to a known KOSHA trigger.

7.4 Fitness-for-Service vs EPC standards

API 579-1 fitness-for-service (FFS) (API, 2020) interacts with EPC design standards (ASME, API design codes) along two independent axes: it **supplements** design-stage codes with a quantitative remaining-life and acceptance methodology for in-service degradation, and it **interfaces** with regulatory obligations because some KOSHA inspection-trigger and documentation requirements are themselves conditioned on FFS-derived quantities (remaining life, corrosion rate). The present system treats FFS as **both** a design-validation input (when applied at the engineering stage to a degraded scenario) **and** an operational assessment (when applied to in-service inspection data). The orchestration is identical in both modes: the calc engine produces an FFS quantity, the four-layer verification confirms numerical consistency, the cross-discipline validator checks for downstream coupling effects, and the KOSHA RAG layer maps the FFS quantity back to any jurisdiction-specific obligation it triggers (e.g., a remaining life below a certain threshold triggers M-69-2012 documentation; a corrosion rate above a certain threshold triggers Article 256 procedural duties). This dual-mode treatment allows compliance detection to follow the asset across its lifecycle without a methodology switch.

7.5 Process-safety contribution

The system contributes to process safety through two pathways. First, the cross-discipline validator proactively identifies coupling hazards that single-domain calculations miss; representative scenarios include excessive piping nozzle loads driving rotating equipment bearing degradation, and civil foundation settlement propagating into rotating equipment alignment failure. Second, as demonstrated in Case 3, the KOSHA RAG layer identifies the corrosion-prevention obligation under Article 256 even when ASME B31.3 and API 570 are fully satisfied. If left unaddressed, unmonitored corrosion progression in a high-temperature, high-pressure carbon-steel line in sour-chloride service may result in leakage or rupture. The system therefore provides a mechanism for automatically detecting the condition of being technically compliant with international standards while remaining legally non-compliant under Korean regulation — a condition that is operationally indistinguishable from “compliant” under current EPC review practice. An anonymised real-plant cryogenic flare-drum data sheet is run end-to-end through the pipeline in Appendix A (VES-REAL-001), confirming that the architecture handles real EPC inputs and surfaces a mandatory KOSHA provision (Article 266 — flare-path block-valve prohibition) for that service.

7.6 Reproducibility and explainability

The architecture prioritises reproducibility and explainability. Every major calculation step carries an explicit standard citation; every regulatory grounding carries a `(reference_code, chunk_id)` traceable to the SQLite index; the golden dataset, the 60-scenario ablation set, the 50-query benchmark, and the orchestration scripts are publicly released, enabling independent replication on commodity hardware with CPU resources and an Ollama installation.

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8. Limitations and Future Work

This study has the following limitations.

1. **Synthetic dataset.** The golden dataset is entirely synthetic; no external validation against real plant operating or design-phase data has been performed. The 220-case accuracy reported in §6.1 constitutes standards-faithful **implementation verification**, not predictive validation against operating-record outcomes.
2. **Screening-level domain outputs.** Some domain outputs represent screening-level assessments and cannot substitute for detailed engineering design review.
3. **Curated retrieval benchmark.** The retrieval benchmark is curated rather than drawn from production user logs; BM25 keyword retrieval may still miss semantically similar Korean documents even with concept-aware synonym expansion and OR fallback. Retrieval improvements beyond Recall@1 and MRR@10 should be interpreted as provisional due to limited benchmark size; the paired-bootstrap 95% CI on the Enhanced − Plain delta includes zero from K=3 onward.
4. **Coupling pair coverage.** The cross-discipline coupling validation is limited to ten predefined domain pairs. Cross-discipline evidence is currently strongest in piping-vessel coupling (22 of 26 ablation detections) and should not yet be overgeneralized to all seven discipline pairs.
5. **Comparison scope.** The reported comparisons are an industry-baseline comparison (“ASME/API pass = compliance complete”) and internal ablations; no third-party head-to-head against a comparable KOSHA-aware multi-discipline system exists because, to our knowledge, no such system has been published.
6. **Twelve omitted law-article rows.** Twelve of 3,102 law-article rows were excluded from the index because their statutory body text was empty in the source API

response. Triage of the twelve (`outputs/kosha_missing_articles_report.md`) finds **four are repealed (삭제) provisions** for which an empty body is expected and the row is correctly excluded from the searchable index; the **remaining eight require a JS-rendered re-fetch** from the National Law Information Center (`law.go.kr/lsScJoRltInfoR.do` returns a single-page-application shell rather than the article body, and the KOSHA Smart-search endpoint cannot retrieve these specific records by `doc_id`). A Playwright-based scraper or migration to an alternate KOSHA OpenAPI endpoint that exposes article bodies by `doc_id` is documented as the engineering follow-up; this is not a research limitation.

7. **Independent expert validation.** The principal author has 12+ years of petrochemical EPC experience; calculation-engine rules and case construction encode that domain practice. Independent validation by a disjoint expert panel — for example, a blind review of system-flagged compliance gaps by external KOSHA PSM auditors — is left to future work and is required before any claim of inter-rater reliability can be made.
8. **Per-discipline accuracy is self-consistency, not external fidelity.** The 1.0000 per-discipline accuracy in Appendix B.1 measures runtime conformance to expected outputs that were themselves generated by the same calculation engine. It is not external fidelity to a separately validated reference. The "implementation verification, not predictive validation" framing of §6.1 applies; field-data validation is documented under future research direction (1).

Future research directions include: (1) **field pilot validation** using real plant data — both design-phase engineering calculations and operating-plant inspection records — to calibrate synthetic thresholds and regulatory trigger conditions; (2) **integration with RBI and HAZOP workflows** to connect system outputs directly with existing PSM documentation; (3) transition to **semantic vector search** using a Korean embedding model in GPU-capable deployment environments; (4) **systematic expansion** of the cross-discipline coupling pair set beyond the current ten pairs; and (5) a dedicated **prospective study** comparing system-flagged regulatory obligations with the findings of independent KOSHA PSM auditors on the same plant package.

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9. Conclusion

This paper has presented an integrated platform combining standards-based calculation engines for seven process plant engineering disciplines, a four-layer hybrid verification model, and a KOSHA regulatory RAG layer applicable across the full plant lifecycle from design through operation. All 220 synthetic golden cases pass end-to-end implementation verification. A 60-scenario cross-discipline ablation shows that enabling the validator increases blocking from 0/60 to 26/60 (+0.4333), with aligned-boundary and aligned-failure subsets rising from 0.0 to 1.0; the detected signal is concentrated in piping-vessel nozzle-margin coupling (22 of 26 detections) and remains exploratory in the other coupling families. A 50-query curated retrieval benchmark shows that enhanced regulatory retrieval improves Recall@1 from 0.4400 to 0.7400 (paired-bootstrap 95% CI [+0.14, +0.48]) and MRR@10 from 0.5744 to 0.7933 (paired CI [+0.09, +0.35]) over plain BM25; Recall@3 and Recall@5 each gain +0.12 with paired CIs that include zero ([−0.02, +0.26]) and are reported for completeness rather than as primary evidence. Three representative cases provide empirical evidence that the KOSHA RAG layer identifies jurisdiction compliance gaps structurally invisible to international-code-only calculation workflows; under the prevailing industry baseline, those same three cases yield 0/3 detections versus 3/3 for the proposed system. Case 3, in which Article 256 corrosion-prevention obligations are detected despite a full ASME/API calculation pass, clearly delineates the structural limitation of calculation-only engineering review.

The framework is positioned as a **compliance co-pilot** for HAZOP, RBI, and digital-twin workflows, not as a replacement for them. Code, golden datasets, the curated benchmark, and the index manifest are publicly released under AGPL-3.0. Future work will focus on field pilot validation with real plant data, prospective comparison with independent KOSHA PSM auditors, transition to semantic vector search, and systematic expansion of the cross-discipline coupling pair set.

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CRediT authorship contribution statement

Yuyong Kim: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Visualization, Writing - original draft, Writing - review and editing.

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Code and Data Availability

Source code, golden datasets, the 60-scenario ablation set, the 50-query curated retrieval benchmark, the SQLite FTS5 index manifest, and the orchestration scripts are publicly released under the AGPL-3.0 license at https://github.com/yuyongkim/did-you-check-kosha (revision tag: `jlp-revised-v2`). A detailed code-to-paper mapping is provided in `docs/publication/CODE_MAP.md`, and the paper source aligned to this release is `docs/publication/submissions_EPC/jlp_revision_2026-05-08/_archive/PAPER_JLP_REVISED_v3.md`. The KOSHA regulatory snapshot was harvested on 2026-02-28 (UTC); the manifest is at `datasets/kosha/manifest.json`.

Acronyms

Acronym	Expansion
ACI	American Concrete Institute
AGPL	Affero General Public License
AISC	American Institute of Steel Construction
API	American Petroleum Institute
ASME	American Society of Mechanical Engineers
BM25	Best Matching 25 (probabilistic ranking function)
CR	Corrosion Rate
EPC	Engineering, Procurement, and Construction
FFS	Fitness-For-Service (per API 579-1/ASME FFS-1)
FTS5	Full-Text Search v5 (SQLite)
HAZOP	Hazard and Operability Study
IEC	International Electrotechnical Commission
IEEE	Institute of Electrical and Electronics Engineers
KOSHA	Korea Occupational Safety and Health Agency
LLM	Large Language Model
MRR	Mean Reciprocal Rank
PSM	Process Safety Management
RAG	Retrieval-Augmented Generation
RBI	Risk-Based Inspection
RL	Remaining Life

Key Definitions

- **Jurisdiction compliance gap** — the set of regulatory obligations imposed by national statute or agency guideline (here, Korean PSM law and KOSHA technical guidelines) that are structurally absent from international engineering code calculations, such that a system may be technically code-compliant while remaining legally non-compliant under the applicable local jurisdiction (§7.1).
- **K-voting** — a Layer-2 verification mechanism in which K=3 calculation paths execute in parallel using deliberately varied numerical strategies, with maximum relative deviation among outputs constrained to ≤ 1%; failure invokes a tiebreaker and, on persistence, raises a `no_consensus` flag (§5.2, §5.4).
- **Coupling validation** — the cross-discipline check across ten predefined domain pairs that fires when an output of one discipline crosses a coupling threshold defined relative to another (e.g., piping reaction force vs vessel nozzle allowable; foundation settlement vs rotating-equipment alignment tolerance) (§5.3).
- **Implementation verification** — confirmation that the system faithfully encodes the standards and rules it claims to implement, evaluated against cases derived from the same standards. Distinguished from **predictive validation**, which would require performance assessment against independently sourced field data (§6.1).
- **Compliance co-pilot** — operating mode in which the system supports, but does not replace, established methods (HAZOP, RBI, Digital Twin) by adding jurisdiction-specific regulatory obligations at four touchpoints (before HAZOP, design review, RBI, digital-twin operation) (§7.2).
- **Mandatory vs Guidance** — regulatory class label attached to each retrieved KOSHA passage; mandatory = provisions of the Act, Enforcement Decree, or *Rules on Occupational Safety and Health Standards*; guidance = KOSHA technical guidelines (§4.3).

Appendix A. Real-Plant Data-Sheet Validation (VES-REAL-001)

A.1 Anonymisation policy and provenance

This appendix exercises the proposed framework against a real engineering data sheet drawn from an operating petrochemical project. **All client, contractor, licensor, personnel, location, project-name, document-ID, and equipment-tag identifiers were stripped before the data sheet was carried into this manuscript or any version-controlled artefact in the repository.** The case is referred to throughout this paper, the supporting scripts, and the JSON/Markdown evidence files only as `VES-REAL-001`. The technical specifications below — material, design and operating pressures and temperatures, dimensions, corrosion allowance, joint-efficiency assumption, fluid specific gravity — are reproduced as supplied by the data sheet and contain no information that could re-identify the source plant, vendor, or design house. No image, watermark, page header, or layout fragment from the source PDF has been embedded in this manuscript. The source PDF itself is retained outside the publication tree (`docs/publication/submissions_EPC/Plant_data/`, not part of the version-controlled paper bundle) and is not redistributed.

A.2 Anonymised vessel specification

The vessel under test is a cryogenic flare knockout drum.

Field	Value
Service	cryogenic flare knockout drum (anonymised real-plant)
Material — shell and heads	SA-240 Type 304/304L (austenitic stainless, dual-cert)
Design pressure	3.5 kg/cm^2(g) ≈ 0.343 MPa(g) **and Full Vacuum**
Design temperature	190 °C / -190 °C (dual range)
Operating P / T	1.5 kg/cm^2(g) / 15.3 °C / -90.5 °C
Inside diameter	5,000 mm (R = 2,500 mm)
Tangent-to-tangent length	20,400 mm
Corrosion allowance	NIL (0 mm)
Joint efficiency	**Not stated on the data sheet — assumed E = 1.0** corresponding to full radiography per ASME Section VIII Div.1 UW-12, conventional for flare-service vessels (assumption explicit; see §A.4)
Liquid specific gravity	0.61

Source of all numbers below: `outputs/real_case_ves001_rag.json` produced by `scripts/run_real_case_ves001_rag.py`. No number in this appendix has been edited by hand.

A.3 Calculation-engine output

The cryogenic side (-190 °C) is outside the input-validation range of the deployed Layer-1 schema (`-50 °C ≤ design_temperature_c ≤ 650 °C`). The full four-layer engine is therefore run on the +190 °C side; the cryogenic side is treated by directly invoking the underlying ASME Section VIII Div.1 UG-27 routine (`src/vessel/calculations.py :: calculate_required_shell_thickness_mm`) with the lowest-tabulated allowable stress (S at 20 °C = 132.0 MPa for SA-240-304L), which is the conventional surrogate for cryogenic shell sizing because allowable stress is not raised below 20 °C. Material toughness for the -190 °C MDMT is governed by ASME Section VIII UHA-51 impact-test rules and is out of scope of the current calculation engine; this is reported honestly here rather than back-filled.

Direct UG-27 results (formula `t = (P·R) / (S·E - 0.6·P) + CA`):

Side	Allowable stress S (MPa)	t_required_shell (mm)	Reverse pressure (MPa)	Reverse deviation (%)
Hot (+190 °C)	118.700	**7.237**	0.3430	$< 1\times10^{-13}$
Cold (-190 °C, S@20 °C)	132.000	6.506	0.3430	0.0000

The hot side governs: **the controlling required shell thickness is 7.237 mm.** The Layer-4 reverse-pressure check returns the design pressure to within machine epsilon (deviation < 0.001 %), as expected for an internally consistent UG-27 forward/reverse pair.

Engine full-run hot-side dimensional metrics (E = 1.0; horizontal drum; 2:1 ellipsoidal heads default):

- diameter = 5,000.0 mm; governing span = 20,400.0 mm; head-depth (default) = 1,250.0 mm
- shell surface area = 320.442 m^2; estimated internal volume = 433.278 m^3
- slenderness L/D = 4.08 (no high-L/D warning)
- screening confidence = `medium`; flags = `{red_flags: [], warnings: ['DATA.VESSEL_DIMENSION_CONTEXT_MISSING']}` — the warning is raised because the data sheet does not state the head depth; the engine substituted its 2:1-ellipsoidal default
- FFS / remaining-life screening fields are *not reported here*: a plausible `current_thickness_mm` was supplied only because the engine schema requires a positive value, and using it would inject a non-data-sheet number into a manuscript table. The script flags this caveat explicitly.

Engine full-run cold-side outcome. As expected, Layer-1 returns the red flag `STD.OUT_OF_SCOPE_APPLICATION` ("Temperature out of range: -190.0") together with the dimension-context warning above. The cold-side response is therefore not used for thickness sizing in this appendix; the directly-computed UG-27 number (6.506 mm) is used instead.

A.4 KOSHA RAG retrieval — both query variants

Two query variants were issued through the same retrieval code path used in the rest of the paper (`src.rag.local_kosha_rag.search_index` against the live SQLite FTS index).

Variant A — `generic_cryogenic_flare_drum_en` (discipline filter = `vessel`). Spec-faithful natural query: *"cryogenic flare knockout drum austenitic stainless steel pressure vessel low temperature service full vacuum 압력용기 저온 플레어"*. Returns 10/10 vessel-discipline hits: **2 mandatory law-article hits + 8 KOSHA technical guides**. Top-3:

Rank	Class	reference_code	Title
1	guidance	`M-111-2015`	압력용기의 용접설계에 관한 기술지침 (Pressure Vessel Weld Design Technical Guide)
2	mandatory	`KOSHA05_안전검사 고시 제9조`	안전검사 고시 제 9 조 정의 (Safety Inspection Notification, Article 9 — Definitions)
3	guidance	`M-69-2012`	압력용기의 잔여수명 평가에 관한 기술지침 (Pressure Vessel Remaining-Life Assessment Technical Guide)

Variant B — `narrow_cryogenic_kr_terms` (no discipline filter). Narrower probe: *"저온 cryogenic 플레어 flare 스테인리스 stainless 압력용기 산업안전보건 process safety"*. Returns 10/10 hits: **1 mandatory law-article hit + 9 KOSHA guides / PSM technical-support documents**. Top-3:

Rank	Class	reference_code	Title
1	guidance	`C-C-86-2026`	공정안전보고서 등의 통합서식 작성방법에 관한 기술지원규정

Rank	Class	reference_code	Title
			(PSM Integrated-Form Preparation Technical-Support Regulation)
2	guidance	`M-69-2012`	압력용기의 잔여수명 평가에 관한 기술지침
3	guidance	`C-C-77-2026`	PSM 시스템의 효과적인 운전 규범에 관한 기술지원규정 (Effective PSM-System Operation Technical-Support Regulation)

The first mandatory hit in Variant B is at **rank 5**:
`KOSHA04_00200000200000400000000000266000` — **제 266 조 차단밸브의 설치 금지** (Article 266, *Rules on Occupational Safety and Health Standards*: prohibition on installing block valves on safety-critical relief / flare paths). This is a directly relevant Korean-jurisdiction obligation for a flare knockout drum; ASME Section VIII Div.1 alone does not surface it.

The full top-10 of each variant, with bm25 scores and snippets, is reproduced verbatim in `outputs/real_case_ves001_rag.md` and `outputs/real_case_ves001_rag.json`.

A.5 What this confirms

The system runs end-to-end on a real EPC vessel data sheet without any code modification: the calculation engine produces a deterministic UG-27 thickness (7.237 mm controlling) that closes its own reverse check to machine epsilon, the engine's dimension layer correctly returns shell volume and L/D screening from the supplied geometry, and the input-validator correctly rejects the cryogenic temperature as out-of-scope rather than silently extrapolating an allowable-stress lookup. KOSHA RAG, queried with a spec-faithful natural query, surfaces a coherent set of jurisdiction-relevant KOSHA technical guides on pressure vessel weld design (`M-111-2015`), remaining-life assessment (`M-69-2012`), pressure-vessel repair (`M-113-2012`), post-weld heat treatment (`M-184-2015`), and thickness-loss risk assessment (`M-109-2012`); the narrow Korean-term probe additionally returns Article 266 of the *Rules on Occupational Safety and Health Standards* — the Korean statutory provision restricting block-valve installation on flare and relief paths — at rank 5, mandatory class. The retrieval is therefore not weak: the spec-faithful variant returns 2 mandatory + 8 vessel-relevant guidance hits, and the narrow variant pulls a flare-relevant mandatory provision into the top-10 along with the PSM integrated-form regulation. We do not claim that the system independently identified Article 266 as the "right" answer for this specific vessel — that is an audit judgement requiring the operating company's PSM dossier — only that the same retrieval pipeline used in §6.5 surfaces the relevant Korean obligation and the relevant KOSHA technical guidance from a real plant data sheet.

A.6 Synthetic-data framing considerations

This real-plant case responds directly to the concern that the §6 evaluation rests on synthetic data alone. What VES-REAL-001 demonstrates is that the pipeline **runs end-to-end on a real EPC vessel data sheet**, that ASME UG-27 governs cleanly, that the engine correctly rejects out-of-scope inputs (the −190 °C cryogenic side returns `STD.OUT_OF_SCOPE_APPLICATION` rather than a fabricated answer), and that KOSHA RAG retrieval surfaces jurisdiction-relevant guidance and at least one mandatory provision (Article 266 on flare-path block-valve prohibition) on a real-plant query. **What VES-REAL-001 does not demonstrate is external accuracy validation against an independently audited reference**: the data sheet supplies design inputs but no independently measured outputs, so the case shows execution and retrieval behaviour, not predictive fidelity. Predictive validation against operating-record outcomes remains future work (§8 Limitation 1).

Reproduction. `python scripts/run_real_case_ves001_rag.py` regenerates `outputs/real_case_ves001_rag.json` and `outputs/real_case_ves001_rag.md`. The KOSHA RAG index used is `datasets/kosha_rag/kosha_local_rag.sqlite3`. No randomness; outputs are deterministic.

Appendix B — Extended validation evidence

This appendix consolidates four supplementary measurements (per-discipline accuracy spread, recall saturation, pipeline latency, retrieval failure inventory) and a reproducibility statement. Every table is regenerated by `scripts/reproduce_all.py` (13/13 scripts pass on a clean run).

B.1 Per-discipline calculation accuracy

A natural follow-up question is whether the system performs unevenly across disciplines. Per-discipline breakdown (`outputs/per_discipline_accuracy.md`):

- Total cases: 220
- Passed cases: 220
- Overall accuracy: 1.0000

- Source dataset: `datasets/golden_standards/*_golden_dataset_v1.json`
- Code path: `scripts/benchmark_all_runtime.py` (re-used).

Per-Discipline Table

Discipline	Cases	Pass	Fail	Accuracy	Primary metric	Mean rel. err	Max rel. err	RF precision	RF recall
piping	50	50	0	1.0000	t_min_mm	0.000000	0.000000	1.0000	1.0000
vessel	30	30	0	1.0000	t_required_shell_mm	0.000000	0.000000	1.0000	1.0000
rotating	30	30	0	1.0000	vibration_mm_per_s	0.000000	0.000000	1.0000	1.0000
electrical	30	30	0	1.0000	transformer_health_index	0.000000	0.000000	1.0000	1.0000
instrumentation	30	30	0	1.0000	pfdavg	0.000000	0.000000	1.0000	1.0000
steel	25	25	0	1.0000	phi_pn_kn	0.000000	0.000000	1.0000	1.0000
civil	25	25	0	1.0000	phi_mn_knm	0.000000	0.000000	1.0000	1.0000

Red-flag Confusion (Per-Discipline). Per-flag, per-case counts: TP = expected and predicted; FP = predicted but not expected; FN = expected but not predicted.

Discipline	TP	FP	FN
piping	29	0	0
vessel	13	0	0
rotating	14	0	0
electrical	24	0	0
instrumentation	28	0	0
steel	26	0	0
civil	26	0	0

Caveat (self-consistency, not external ground-truth fidelity). The expected outputs in `datasets/golden_standards/*.json` were generated by the same calculation engine that the runtime now reproduces. The 1.0000 uniform accuracy therefore measures *implementation self-consistency* — the engine's behaviour at runtime matches its behaviour at dataset-generation time — not external fidelity to a separately validated reference. This caveat is consistent with the "implementation verification, not predictive validation" framing of §6.1 and is restated as Limitation 8 in §8.

B.2 Recall@K extension (K = 1..10)

Per-K table from `outputs/rag_retrieval_extended.md` (Dataset: `datasets/kosha_rag/rag_eval_queries.json`; queries: 50; bootstrap draws: 1000, seed=20260508):

Point Estimates per K

K	Plain	Enhanced	Delta (E - P)
1	0.4400	0.7400	+0.3000
2	0.6000	0.8200	+0.2200
3	0.7400	0.8600	+0.1200
4	0.7400	0.8600	+0.1200
5	0.7400	0.8600	+0.1200
6	0.7600	0.8600	+0.1000
7	0.7600	0.8600	+0.1000
8	0.7600	0.8600	+0.1000
9	0.8000	0.8600	+0.0600
10	0.8000	0.8600	+0.0600

Paired Difference (Enhanced – Plain) per K

K	Mean	2.5%	97.5%	CI excludes 0?

K	Mean	2.5%	97.5%	CI excludes 0?
1	+0.3031	+0.1400	+0.4800	yes
2	+0.2236	+0.0600	+0.3800	yes
3	+0.1216	-0.0200	+0.2600	no
4	+0.1216	-0.0200	+0.2600	no
5	+0.1216	-0.0200	+0.2600	no
6	+0.1020	-0.0200	+0.2200	no
7	+0.1020	-0.0200	+0.2200	no
8	+0.1020	-0.0200	+0.2200	no
9	+0.0609	-0.0600	+0.1800	no
10	+0.0609	-0.0600	+0.1800	no

The Enhanced curve saturates at 0.8621 from K = 3; the Plain curve continues to rise. Paired-bootstrap 95% CI on the Enhanced − Plain delta excludes zero only at K∈{1, 2}. The headline retrieval claim of the paper is therefore narrowed to Recall@1 and MRR@10 (§6.5).

B.3 Pipeline end-to-end latency

Median latency per stage on a commodity AMD x86-64 CPU under Python 3.13 (`outputs/pipeline_latency.md`; Platform: Windows 10 (AMD64); CPU: AMD64 Family 25 Model 33 Stepping 2; repetitions per stage: 5):

Per-Discipline Layered Stack (Layers 1–4 Combined, service.evaluate)

Discipline	Case ID	Criticality	Median (ms)	p95 (ms)	Min (ms)	Max (ms)
piping	PIP-GOLD-001	normal	0.06	0.19	0.05	0.19
vessel	VES-GOLD-001	normal	0.06	0.13	0.05	0.13
rotating	ROT-GOLD-001	normal	0.04	0.10	0.04	0.10
electrical	ELE-GOLD-001	normal	0.04	0.09	0.03	0.09
instrumentation	INS-GOLD-	normal	0.08	0.15	0.06	0.15

Discipline	Case ID	Criticality	Median (ms)	p95 (ms)	Min (ms)	Max (ms)
	001					
steel	STL-GOLD-001	normal	0.03	0.07	0.03	0.07
civil	CIV-GOLD-001	normal	0.03	0.08	0.03	0.08

Piping Per-Layer Breakdown (case PIP-GOLD-001):

Layer	Median (ms)	p95 (ms)	Min (ms)	Max (ms)
layer1_input_validation	0.002	0.004	0.001	0.004
layer2_k_voting_3candidates_plus_consensus	0.017	0.020	0.015	0.020
layer3_physics_and_standards	0.002	0.005	0.002	0.005
layer4_reverse_checks	0.003	0.005	0.002	0.005

Cross-Discipline Validator (one paired call across 7 disciplines): median 0.028 ms, p95 0.045 ms, min/max 0.021 / 0.045 ms.

KOSHA RAG Retrieval (top-10): query `pressure vessel remaining life assessment`, hit count 4, median 1.151 ms, p95 3.351 ms, min/max 1.018 / 3.351 ms.

KOSHA RAG Generation (Qwen via Ollama): model `qwen3:4b`, prompt 2,054 chars, median 41,725.0 ms, p95 62,277.2 ms, min/max 31,919.7 / 62,277.2 ms.

End-to-End Median. Composition: piping `service.evaluate` median + cross-discipline validator median + RAG retrieval median (+ RAG generation median). End-to-end median: **41,726.22 ms**.

The deterministic stack (input validation, K-voting, physics/code compliance, reverse verification, cross-discipline validator, BM25 retrieval) completes in approximately 1.2 ms per case. The Qwen-family generation step on the on-premises Ollama server adds approximately 42 s median; this dominates end-to-end latency when full LLM grounding is requested. K-voting (L2) accounts for approximately 70% of within-engine layered latency, consistent with its use of three deliberately varied calculation paths.

B.4 Retrieval failure inventory

Honest disclosure of where retrieval still misses on the 50-query curated benchmark (`outputs/retrieval_failure_inventory.md`; Enhanced FTS, K=5 threshold; failure rule: `failure := first_relevant_rank > 5 OR no relevant hit in top-10`; failure count 7/50 = 0.1400):

Diagnosis	Count
paraphrase_too_loose	6
regulatory_class_mismatch	1

Diagnosis category definitions: `paraphrase_too_loose` — the query phrasing diverges from the indexed chunk's vocabulary beyond what the synonym expansion bridges; `regulatory_class_mismatch` — the retrieval favours a guidance-class hit when the target is a mandatory provision (or vice versa). Six of seven failures fall in `paraphrase_too_loose`, concentrated in the C-C-23 RBI, B-M-18 piping life, C-C-75 corrosion risk, and Article 256 corrosion-prevention groups. This single dominant failure mode is an honest input for future work on a hybrid lexical + semantic retriever (§8 future research direction (3)).

B.5 Reproducibility

`scripts/reproduce_all.py` regenerates every figure, table, and `outputs/*.md` cited in this manuscript with a single command from a clean repository state. End-to-end run on a clean checkout: 13/13 scripts complete successfully (`python scripts/reproduce_all.py`). The KOSHA-corpus rebuild (network + 200 MB SQLite) is intentionally excluded from the runner and documented separately under `scripts/sync_kosha_corpus.py`.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the author used OpenAI Codex (GPT-5 based coding assistant) to help restructure prose, normalise journal-formatting details, and prepare supporting submission files. After using this tool, the author reviewed and edited the content as needed and takes full responsibility for the content of the article.

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